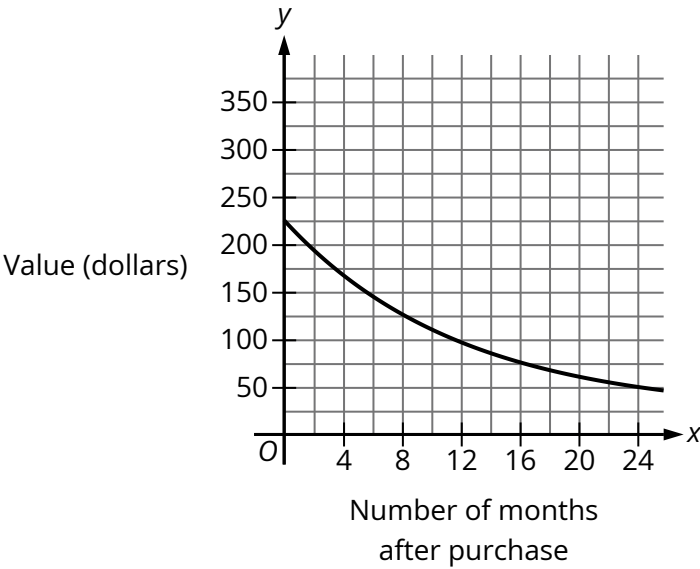


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question



The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y -intercept of the graph in this context?

Answer

- A. The estimated value of the tablet was **\$225** when it was purchased.
- B. The estimated value of the tablet **24** months after it was purchased was **\$225**.
- C. The estimated value of the tablet had decreased by **\$225** in the **24** months after it was purchased.
- D. The estimated value of the tablet decreased by approximately **2.25%** each year after it was purchased.

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph shown gives the estimated value y , in dollars, of a tablet as a function of the number of months since it was purchased, x . The y -intercept of a graph is the point at which the graph intersects the y -axis, or when x is 0. The graph shown intersects the y -axis at the point **(0, 225)**. It follows that **0** months after the tablet was purchased, or when the tablet was purchased, the estimated value of the tablet was **225** dollars. Therefore, the best interpretation of the y -intercept is that the estimated value of the tablet was **\$225** when it was purchased.

Choice B is incorrect. The estimated value of the tablet **24** months after it was purchased was **\$50**, not **\$225**.

Choice C is incorrect. The estimated value of the tablet had decreased by **\$225 — \$50**, or **\$175**, not **\$225**, in the **24** months after it was purchased.

Choice D is incorrect and may result from conceptual errors.